CS182: Introduction to Machine Learning
Lecture 06: Parameter Estimation

Worked Solutions and Marking Guide

Scope: Parameter_Estimation.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Quick key — Section A

1	2	3	4	5	6
F	F	F	T	F	F

7	8	9	10	11	12
F	T	F	F	F	T

Quick key — Section B

13	14	15	16	17	18
ABD	ACD	AD	ABD	AB	AC

19	20	21	22	23	24
CD	B	C	BC	BCD	CD

Section A. True/False — Explanations

1. False. Centering gives rank at most $d-1$, so it is singular. Positive diagonal entries do not imply full rank because the columns may be linearly dependent.
Source: PDF pp.59,63,71,74; supplementary centered-rank argument.

2. False. The posterior mean is optimal under squared error loss. Other losses can lead to other optimal estimates; for example, absolute error uses a posterior median.
Source: PDF pp.24-25; supplementary absolute-loss contrast.

3. False. Zero covariance only excludes linear association. Joint normality is a special case in which zero covariance implies independence.
Source: PDF pp.58,65-66.

4. True. The common quadratic term and log determinant cancel across classes. Positive unequal priors change only the offsets. Distinct class means give a hyperplane equality boundary; identical means give a constant difference, with either no equality boundary or ties everywhere.
Source: PDF pp.70,75-76.

5. False. The plug-in variance is the observation variance σ^2. Full prediction also includes posterior uncertainty in the mean, giving $\sigma^2 + \mathrm{Var}(\theta \mid D)$. Normality makes posterior mean and mode coincide; it does not remove uncertainty.
Source: PDF pp.23-28; supplementary evaluated predictive variance.

6. False. If n values were observed and $m > 0$ were missing, the filled entries add zero centered squared deviation. The centered sum of squares stays the same but the denominator becomes $n+m$, shrinking a nonzero ML variance by $n/(n+m)$.
Source: PDF pp.59-60; supplementary imputation calculation.

7. False. MSE is squared bias plus variance. A biased estimator can have enough smaller variance to obtain lower MSE.
Source: PDF pp.17-19.

8. True. The MLE is its count divided by N. Assigning positive mass to an unseen category would remove mass from observed categories and reduce their product likelihood. An imposed positive-probability restriction or a prior changes this conclusion.
Source: PDF pp.14-15; supplementary boundary condition.

9. False. The likelihood is $p(D \mid \theta)$ viewed as a function of θ, with D fixed. It is normalized over possible data for each parameter, not generally over the parameter. A posterior over θ requires a prior and a normalizing evidence.
Source: PDF pp.9,22.

10. False. Different diagonal variances leave different coefficients of x_j^2 in the score differences. Conditional independence removes cross-products, but shared covariance is what cancels all quadratic terms.
Source: PDF pp.70-72,75,78.

11. False. Its expectation is $(N-1)\sigma^2/N$, so it has downward bias. The denominator $N-1$ gives the usual unbiased estimator for $N > 1$.
Source: PDF pp.16,19.

12. True. The model $w_0 + w_1 x + w_2 x^2 + w_3 \sin x$ is linear in the coefficients w, even though its dependence on the original input is nonlinear.
Source: PDF pp.42,44,90.

Section B. Multiple Response — Explanations

13. Correct set: A, B, D.

(A) **Correct.** The absent category contributes no observations, so its likelihood factor is 1; the notation $0 \log 0 = 0$ represents this limiting convention.

(B) **Correct.** The categorical likelihood is maximized at relative category counts.

(C) **Incorrect.** The probabilities must sum to 1. Independent unconstrained optimization would not produce a normalized categorical model.

(D) **Correct.** The maximum has $p_2 = 0$, outside that assumed interior. Positive-count probabilities can be derived on their face of the simplex.

Source: PDF pp.14-15; supplementary zero-count handling.

14. Correct set: A, C, D.

(A) **Correct.** The log-density contribution is $-[r^t - g(x^t)]^2/(2\sigma_t^2)$, up to terms independent of the mean parameters.

(B) **Incorrect.** It can only be dropped when it does not depend on the fitted parameters; the slides use that assumption explicitly.

(C) **Correct.** The parameter-dependent log likelihood is a negative positive-scaled sum of squared residuals.

(D) **Correct.** The design columns may be x_j^2, interactions, or $\sin x_j$; linearity here concerns the unknown coefficients.

Source: PDF pp.38-44,88-90; supplementary known unequal-variance extension.

15. Correct set: A, D.

(A) **Correct.** Its denominator is $N - 1 = 4$, giving $20/4 = 5$.

(B) **Incorrect.** That denominator corrects finite-sample bias; the likelihood optimum uses N.

(C) **Incorrect.** The standard deviation MLE is the positive square root of the variance MLE: $\sqrt{4} = 2$. The number five is the unbiased variance estimate.

(D) **Correct.** The Gaussian maximum-likelihood variance is the centered sum of squares divided by N: $20/5 = 4$.

Source: PDF pp.16-19.

16. Correct set: A, B, D.

(A) **Correct.** The quadratic form is $3^2/9 + 1^2/1 = 2$.

(B) **Correct.** A diagonal Gaussian covariance factorizes the joint normal density into univariate normal densities.

(C) **Incorrect.** It is $3^2 + 1^2 = 10$; Mahalanobis distance rescales the coordinates.

(D) **Correct.** The positive covariance aligns larger simultaneous deviations along the rising direction.

Source: PDF pp.64-68.

Section B. Multiple Response — Explanations

17. Correct set: A, B.

(A) **Correct.** The posterior precision is $1/4 + 3/9 = 7/12$, so its variance is $12/7$.

(B) **Correct.** The precision-weighted numerator is $1/4 + (3/9)4 = 19/12$; dividing by $7/12$ gives $19/7$.

(C) **Incorrect.** The Gaussian posterior mode equals its posterior mean $19/7$; the value four is the likelihood maximizer before incorporating the prior.

(D) **Incorrect.** It is $9 + 12/7 = 75/7$; variance 9 belongs to the plug-in distribution.

Source: PDF pp.22-28; supplementary predictive evaluation.

18. Correct set: A, C.

(A) **Correct.** Expanding each term gives $x_j \log[p_{ij}/(1 - p_{ij})] + \log(1 - p_{ij})$, plus the log prior.

(B) **Incorrect.** For a zero-valued Bernoulli feature, the factor is $1 - p_{ij}$, which is essential to the likelihood.

(C) **Correct.** They are $0.6(1 - 0.5)(0.8) = 0.24$ and $0.4(1 - 0.9)(0.3) = 0.012$.

(D) **Incorrect.** The value 0.24 is an unnormalized joint score. Normalize both scores to obtain $0.24/(0.24 + 0.012) = 20/21$.

Source: PDF p.85.

19. Correct set: C, D.

(A) **Incorrect.** For uncertain parameters it averages likelihood over a normalized parameter prior: $p(D \mid M) = \int p(D \mid \theta, M)p(\theta \mid M)\, d\theta$. A maximum is a different operation.

(B) **Incorrect.** Validation estimates generalization by evaluating data unused in that predictor's training.

(C) **Correct.** The objective is data error plus λ times a complexity penalty; changing λ changes this tradeoff.

(D) **Correct.** Bayes' rule gives $p(M \mid D) \propto p(D \mid M)p(M)$.

Source: PDF pp.22-23,47,51-53; supplementary evidence integral.

20. Correct set: B.

(A) **Incorrect.** A maximizer can occur on the boundary; derivative-zero conditions only locate interior candidates.

(B) **Correct.** The likelihood is p^5, which increases up to the right endpoint.

(C) **Incorrect.** Its derivative is $5/p > 0$, so no interior stationary point exists.

(D) **Incorrect.** The likelihood p^5 is strictly increasing throughout the open interval. Its supremum is approached as $p \to 1$ and is never attained; $4/5$ is not a maximizer.

Source: PDF pp.9-11; supplementary boundary analysis.

Section B. Multiple Response — Explanations

21. Correct set: C.

(A) **Incorrect.** They sum to 1, leaving $K-1=2$ free parameters. Means additionally require $Kd=12$ parameters.

(B) **Incorrect.** All coordinate variances share one scalar s^2, so only one covariance parameter is free.

(C) **Correct.** Each symmetric 4×4 matrix has $4(5)/2=10$ independent entries; three require 30.

(D) **Incorrect.** A covariance matrix is symmetric. Its $4(5)/2=10$ independent entries are fewer than its 16 displayed matrix entries.

Source: PDF pp.14-15,74-75,78,80,82.

22. Correct set: B, C.

(A) **Incorrect.** The lecture's bias-variance discussion is a common tendency, not a universal strict monotonicity result.

(B) **Correct.** Two centered square expansions give $\sigma^2(x)+[E_D g_D(x)-f(x)]^2+\mathrm{Var}_D(g_D(x))$.

(C) **Correct.** Both prediction bias and training-sample variance vanish, leaving response noise.

(D) **Incorrect.** Validation estimates generalization from a finite sample and therefore has sampling uncertainty.

Source: PDF pp.48-53,82; supplementary precise probabilistic conditions.

23. Correct set: B, C, D.

(A) **Incorrect.** Here $7/16<2/3$, so the biased maximum-likelihood variance has lower MSE.

(B) **Correct.** The sample mean has expectation μ and variance σ^2/N; Chebyshev's inequality makes its deviation probability tend to zero.

(C) **Correct.** Its bias is $3/4-1=-1/4$, its variance is $6/16$, and their MSE sum is $1/16+6/16=7/16$.

(D) **Correct.** It is unbiased and has variance $6/9=2/3$.

Source: PDF pp.17-20; supplementary supplied Gaussian residual moments.

24. Correct set: C, D.

(A) **Incorrect.** It makes full rank possible, but the centered observations may still occupy a lower-dimensional subspace.

(B) **Incorrect.** Its rank is at most $\min(d,N-K)=\min(5,9)=5$; pooled within-class directions can span all five dimensions.

(C) **Correct.** Its rank is at most $N_i-1=3<5$.

(D) **Correct.** If all estimated coordinate variances are positive, the diagonal matrix is invertible even though the centered sample vectors span fewer than d dimensions.

Source: PDF pp.59,63,71,74-75,78; supplementary rank constraints.

Section C. Proofs and Formula Derivations

25. Discrete likelihoods, constraints, and naive Bayes **[10 marks]**

(a) [3 marks] The likelihood is $L(p) = p^s(1-p)^{N-s}$. For $0 < s < N$,

$$\ell'(p) = \frac{s}{p} - \frac{N-s}{1-p} = 0 \quad \Longrightarrow \quad \hat{p} = s/N.$$

The second derivative $-s/p^2 - (N-s)/(1-p)^2 < 0$ gives strict concavity on the interior, and the likelihood is zero at both endpoints, so this is the unique global maximum. If $s = 0$, $L = (1-p)^N$ is maximized at 0; if $s = N$, $L = p^N$ is maximized at 1. Marking: likelihood/derivative (1), interior global maximum (1), both boundary cases (1).

(b) [4 marks] The log likelihood is $\ell(p) = \sum_k n_k \log p_k$, with $p_k \geq 0$ and $\sum_k p_k = 1$. First suppose all counts are positive. The Lagrangian $\ell + \eta(1 - \sum_k p_k)$ gives $n_k/p_k = \eta$; summing $p_k = n_k/\eta$ yields $\eta = N$, hence $\hat{p}_k = n_k/N$. Strict concavity on the positive-count simplex proves global maximality. For zero-count categories, let their total probability be z. Write the observed-category probabilities as $p_k = (1-z)q_k$, with the q_k summing to 1. Then

$$\ell(p) = N \log(1-z) + \sum_{n_k > 0} n_k \log q_k.$$

This is maximized at $z = 0$; any positive mass on unseen categories reduces the likelihood. Applying the previous derivation on the observed categories gives the same formula for all k, including zeros. Marking: constrained objective (1), multiplier derivation (1), global maximum (1), zero-count argument (1).

(c) [3 marks] For class C_i,

$$p(x \mid C_i) = \prod_{j=1}^{d} p_{ij}^{x_j} (1 - p_{ij})^{1-x_j}.$$

With $N_i > 0$ class-i observations, the log likelihood separates over j, giving $\hat{p}_{ij} = (1/N_i) \sum_{t:C^t=C_i} x_j^t$. The class prior MLE is N_i/N. For $0 < p_{ij} < 1$,

$$g_i(x) = \log P(C_i) + \sum_j \log(1 - p_{ij}) + \sum_j x_j \log \frac{p_{ij}}{1 - p_{ij}},$$

which is affine in x. A zero or unit estimate requires handling zero likelihoods or infinite logs; finite coefficients cannot simply be assumed. Marking: likelihood (1), MLEs (1), affine expansion and boundary qualification (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.9-15,85-86; supplementary boundary analysis.

Section C. Proofs and Formula Derivations

26. Gaussian variance and two kinds of bias-variance decomposition [10 marks]

(a) [3 marks] For each $v > 0$, the identity shows that the squared-deviation term is minimized at $\mu = \bar{X}$. The profile log likelihood is

$$\ell(v) = \text{constant} - \frac{N}{2}\log v - \frac{T}{2v}, \quad \ell'(v) = \frac{T - Nv}{2v^2}.$$

It increases for $v < T/N$ and decreases for $v > T/N$, so $\hat{v} = T/N$ is the global maximum. If $T = 0$, $\ell(v) = \text{constant} - (N/2)\log v \to +\infty$ as $v \downarrow 0$. No maximizer exists within the ordinary positive-variance model. Marking: profiling (1), derivative/global maximum (1), degeneracy (1).

(b) [2 marks] The identity $T = \sum_t (X_t - \mu)^2 - N(\bar{X} - \mu)^2$ gives

$$E[T] = Nv - N\,\mathrm{Var}(\bar{X}) = Nv - N(v/N) = (N-1)v.$$

Thus $E[T/N] = (N-1)v/N$, with bias $-v/N$. For $N > 1$, $T/(N-1)$ is unbiased. Marking: expectation calculation using v/N (1), bias and correction (1).

(c) [2 marks] Write $d - \theta = (d - E[d]) + (E[d] - \theta)$. Squaring and taking expectations gives the variance plus squared bias, because $E[d - E[d]] = 0$ makes the cross term vanish. Removing bias does not control variance; a biased estimator can have smaller variance by more than its added squared bias. Marking: centered expansion/cross term (1), identity and comparison implication (1).

(d) [3 marks] First condition on D:

$$E_Y[(Y - g_D(x))^2 \mid x, D] = \sigma^2(x) + (f(x) - g_D(x))^2,$$

since $E[Y - f(x) \mid x, D] = 0$ by independent test sampling. Let $\bar{g}(x) = E_D[g_D(x)]$. Expanding $f - g_D = (f - \bar{g}) + (\bar{g} - g_D)$ and averaging over D removes the second cross term. Therefore

$$E_{D,Y}[(Y - g_D(x))^2 \mid x] = \sigma^2(x) + [f(x) - \bar{g}(x)]^2 + \mathrm{Var}_D(g_D(x)).$$

This bias concerns the average fitted prediction relative to $f(x)$; parameter-estimator bias instead compares $E[d]$ with a fixed parameter θ. Marking: response-noise decomposition (1), training-set decomposition (1), final identity and distinction (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.16-20,48-49; supplementary degenerate-sample qualification.

Section C. Proofs and Formula Derivations

27. A Gaussian prior is more than a point estimate **[10 marks]**

(a) [4 marks] Ignoring factors independent of θ, the posterior kernel is

$$\exp\left\{-\frac{1}{2}\left[\frac{\sum_t (X_t-\theta)^2}{\sigma^2}+\frac{(\theta-m_0)^2}{\tau^2}\right]\right\}.$$

The coefficient of θ^2 is $A = N/\sigma^2 + 1/\tau^2$; the linear coefficient is $-2B$, where $B = \sum_t X_t/\sigma^2 + m_0/\tau^2$. Since $A\theta^2 - 2B\theta = A(\theta - B/A)^2 - B^2/A$, the normalized posterior is

$$\theta \mid D \sim N(m_N, v_N), \qquad v_N = A^{-1}, \quad m_N = v_N\left(\frac{N\bar{X}}{\sigma^2}+\frac{m_0}{\tau^2}\right).$$

Thus posterior precision is the sum of data and prior precisions, and the mean is their normalized precision-weighted average. Marking: kernel (1), quadratic coefficients/completion (1), variance (1), mean/weights (1).

(b) [3 marks] Conditional on the data, represent $X_* = \theta + \varepsilon_*$, where $\theta \sim N(m_N, v_N)$ and $\varepsilon_* \sim N(0, \sigma^2)$ are independent. Hence

$$p(X_* \mid D) = \int p(X_* \mid \theta) p(\theta \mid D)\, d\theta = N(m_N, \sigma^2 + v_N).$$

Equivalently, the laws of total expectation and variance give mean m_N and variance $\sigma^2 + v_N$, and Gaussian convolution identifies the normal distribution. Plugging in $\theta = m_N$ gives $N(m_N, \sigma^2)$, omitting posterior parameter uncertainty $v_N > 0$. Marking: posterior averaging/convolution (1), predictive distribution (1), plug-in comparison (1).

(c) [3 marks] For any point estimate a,

$$E[(\theta - a)^2 \mid D] = v_N + (m_N - a)^2,$$

so the unique squared-loss minimizer is $a = m_N$. The normal posterior also has its mode at m_N, so MAP and the posterior mean agree here. As $\tau^2 \to \infty$, prior precision vanishes: $m_N \to \bar{X}$ and $v_N \to \sigma^2/N$. This variance remains positive for finite N; full prediction has limiting variance $\sigma^2(1 + 1/N)$, whereas the plug-in variance remains σ^2. Marking: squared-loss argument (1), MAP relationship (1), diffuse limits and predictive consequence (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.22-28; supplementary evaluated predictive distribution.

Section C. Proofs and Formula Derivations

28. Gaussian discriminants and when covariance estimates can be used [10 marks]

(a) [3 marks] Taking logs and removing only the common $-d\log(2\pi)/2$ gives

$$g_i(x) = -\frac{1}{2}\log|S_i| - \frac{1}{2}(x-m_i)^\top S_i^{-1}(x-m_i) + \log\pi_i.$$

Using symmetry of S_i^{-1}, expand the quadratic form to obtain

$$W_i = -\tfrac{1}{2}S_i^{-1}, \quad w_i = S_i^{-1}m_i, \quad b_i = -\tfrac{1}{2}m_i^\top S_i^{-1}m_i - \tfrac{1}{2}\log|S_i| + \log\pi_i.$$

Different S_i^{-1} generally leave quadratic terms in pairwise score differences, producing QDA boundaries. Marking: log density (1), expansion/coefficients (1), boundary implication (1).

(b) [3 marks] The terms $-x^\top S^{-1}x/2$ and $-\log|S|/2$ are common and cancel. Equivalent scores are

$$\tilde{g}_i(x) = m_i^\top S^{-1}x - \tfrac{1}{2}m_i^\top S^{-1}m_i + \log\pi_i.$$

A boundary between distinct means satisfies

$$(m_i-m_j)^\top S^{-1}x - \tfrac{1}{2}(m_i^\top S^{-1}m_i - m_j^\top S^{-1}m_j) + \log(\pi_i/\pi_j) = 0,$$

a hyperplane. For equal priors, maximizing the original scores minimizes $(x-m_i)^\top S^{-1}(x-m_i)$, a squared Mahalanobis distance. This agrees with ordinary Euclidean nearest mean when $S = s^2 I$ with common $s^2 > 0$. A shared diagonal matrix generally gives unequal coordinate weights. Marking: cancellation/affine score (1), boundary (1), metric interpretations (1).

(c) [4 marks] A symmetric $d \times d$ covariance has $d(d+1)/2$ free entries. Separate matrices need $Kd(d+1)/2$, shared full covariance needs $d(d+1)/2$; both also need Kd mean entries and $K-1$ free priors. Form $Z_i \in \mathbb{R}^{N_i \times d}$ with centered observations as rows. Its rows sum to zero, so $\operatorname{rank}(Z_i) \le N_i - 1$. Since $S_i = Z_i^\top Z_i/N_i$,

$$\operatorname{rank}(S_i) \le \min(d, N_i - 1).$$

Thus $N_i \ge d+1$ is necessary for a class covariance to be invertible. Stacking all Z_i gives Z and

$$S = \frac{1}{N}Z^\top Z, \qquad \operatorname{rank}(S) \le \min\left(d, \sum_i (N_i - 1)\right) = \min(d, N-K).$$

Hence $N - K \ge d$ is necessary for pooled invertibility. These inequalities only allow full rank: the centered observations must also span $\mathbb{R}^d$. Redundant features or samples confined to a subspace still cause singularity, invalidating the ordinary inverse/determinant density formula. Marking: counts (1), class rank/condition (1), pooled rank/condition (1), spanning and insufficiency (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.59,63,70-82; supplementary centered-rank conditions.

Section D. Applications

29. Estimating a Gaussian classifier from rectangular samples [12 marks]

(a) [4 marks] Both classes contain four of eight observations, giving $\hat{\pi}_1 = \hat{\pi}_2 = 1/2$. The means are $m_1 = (0,0)^\mathsf{T}$, $m_2 = (2,2)^\mathsf{T}$. Within either class, centered vectors are $(\pm 1, \pm 2)$ with all four sign combinations. Their centered outer products sum to $\mathrm{diag}(4,16)$, since the cross-products cancel. Dividing by 4 gives

$$S_1 = S_2 = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}.$$

The pooled estimate $(1/2)S_1 + (1/2)S_2$ is the same matrix. Marking: priors/means (1), centered outer-product computation (2), pooled estimate (1).

(b) [3 marks] Here $S^{-1} = \mathrm{diag}(1, 1/4)$. With equal priors,

$$g_2 - g_1 = (m_2 - m_1)^\mathsf{T} S^{-1} x - \tfrac{1}{2}(m_2^\mathsf{T} S^{-1} m_2 - m_1^\mathsf{T} S^{-1} m_1) = 2x_1 + \tfrac{1}{2}x_2 - \tfrac{5}{2}.$$

Thus the boundary is $4x_1 + x_2 = 5$, with C_2 preferred where $4x_1 + x_2 > 5$. The separately fitted covariances happen to be equal, so the QDA quadratic terms cancel and produce this same classifier. Marking: score difference (1), boundary/side (1), QDA comparison (1).

(c) [2 marks] The squared Mahalanobis distances are $0.5^2 + 2^2/4 = 1.25$ to m_1 and $(-1.5)^2 + 0^2/4 = 2.25$ to m_2, so LDA chooses C_1. Euclidean distances squared are 4.25 and 2.25, so ordinary nearest mean chooses C_2. LDA discounts deviations in the second coordinate because its within-class variance is four times as large. Marking: distances and choices (1), covariance-based explanation (1).

(d) [3 marks] The covariance counts are $2[2(3)/2] = 6$, $2(3)/2 = 3$, $d = 2$, and 1, respectively. The fitted covariance is shared diagonal with unequal coordinate variances, so the shared diagonal model matches it exactly; shared full and separate full also contain it. A shared spherical matrix cannot equal $\mathrm{diag}(1,4)$. Here each $N_i = 4 \geq d + 1 = 3$, and $N - K = 8 - 2 = 6 \geq d = 2$. These are necessary conditions only; the directly computed determinant $1(4) = 4 > 0$ establishes invertibility for this sample. Marking: counts (1), exact covariance assumptions (1), sample-size checks versus determinant (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.59,63-64,70-82; supplementary numerical sample.

Section D. Applications

30. Gaussian regression, prediction error, and model selection [12 marks]

(a) [3 marks] The joint likelihood factors as $\prod_t p(y^t \mid x^t, w, \sigma^2)p(x^t)$. Its log is

$$-\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - Xw\|^2 + \sum_t \log p(x^t),$$

where row t of X is $(1, x^t)$. At any fixed positive σ^2, maximizing in w is equivalent to minimizing $E(w) = \frac{1}{2}\|y - Xw\|^2$. Differentiating gives $\nabla E = X^\mathsf{T}(Xw - y)$, so every optimum satisfies $X^\mathsf{T}Xw = X^\mathsf{T}y$. The inverse expression requires full column rank. Marking: likelihood and parameter-independent input term (1), squared-error equivalence (1), normal equations/rank condition (1).

(b) [3 marks] The sums are $N = 4$, $\sum x = 2$, $\sum x^2 = 6$, $\sum y = 8$, and $\sum xy = 12$. Thus

$$\begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}\begin{pmatrix} \hat{w}_0 \\ \hat{w}_1 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}.$$

The determinant is 20, and the solution is $\hat{w}_0 = 1.2, \hat{w}_1 = 1.6$. The residuals $y - \hat{y}$ are $0.4, -0.2, -0.8, 0.6$, giving SSE $= 0.16 + 0.04 + 0.64 + 0.36 = 1.20$. The Gaussian variance MLE is SSE$/N = 0.30$; a degrees-of-freedom correction addresses unbiasedness rather than likelihood maximization. Marking: normal system/solution (1), residuals and SSE (1), variance MLE (1).

(c) [3 marks] The squared biases are $(1.7 - 2)^2 = 0.09$, $(1.9 - 2)^2 = 0.01$, and 0. Adding variance and irreducible noise gives

$$E_1 = 0.25 + 0.09 + 0.09 = 0.43, \quad E_2 = 0.25 + 0.01 + 0.12 = 0.38, \quad E_3 = 0.25 + 0 + 0.50 = 0.75.$$

Degree 2 is best at this input. Degree 3 removes this bias but pays a larger training-sample variance. Marking: squared biases (1), three total errors (1), choice and explanation (1).

(d) [3 marks] The equally weighted validation means are $1.2, 0.8, 1.2$, so cross-validation selects degree 2. For each fold, fit only on the other folds and score the held-out observations; after choosing the degree, refit that model on the available training data. A final test set should remain unused during tuning. More flexible nested polynomial models can reduce training SSE while fitting sampling noise, so their training-error ordering does not determine future error. Finite-fold validation is an estimate rather than a guarantee of the population ranking. Marking: fold averages/choice (1), correct separation and refit (1), training-versus-generalization explanation (1).

Marking: use the displayed subpart allocations; award credit for the logical or numerical intermediate steps.

Source: PDF pp.38-44,47-53,88-90; supplementary numerical model comparison.

Lecture Alignment and Coverage

Primary source: `Parameter_Estimation.pdf` in `E:\CS182\2026fall\lecture`. All PDF pages count the cover as page 1. Additional arguments are marked as supplementary derivations when appropriate. These are original practice questions, not copied homework questions.

T/F

1 PDF pp.59,63,71,74; supplementary centered-rank argument.

2 PDF pp.24-25; supplementary absolute-loss contrast.

3 PDF pp.58,65-66.

4 PDF pp.70,75-76.

5 PDF pp.23-28; supplementary evaluated predictive variance.

6 PDF pp.59-60; supplementary imputation calculation.

7 PDF pp.17-19.

8 PDF pp.14-15; supplementary boundary condition.

9 PDF pp.9,22.

10 PDF pp.70-72,75,78.

11 PDF pp.16,19.

12 PDF pp.42,44,90.

Multiple response

13 PDF pp.14-15; supplementary zero-count handling.

14 PDF pp.38-44,88-90; supplementary known unequal-variance extension.

15 PDF pp.16-19.

16 PDF pp.64-68.

17 PDF pp.22-28; supplementary predictive evaluation.

18 PDF p.85.

19 PDF pp.22-23,47,51-53; supplementary evidence integral.

20 PDF pp.9-11; supplementary boundary analysis.

21 PDF pp.14-15,74-75,78,80,82.

22 PDF pp.48-53,82; supplementary precise probabilistic conditions.

23 PDF pp.17-20; supplementary supplied Gaussian residual moments.

24 PDF pp.59,63,71,74-75,78; supplementary rank constraints.

Proofs and applications

25 PDF pp.9-15,85-86; supplementary boundary analysis.

26 PDF pp.16-20,48-49; supplementary degenerate-sample qualification.

27 PDF pp.22-28; supplementary evaluated predictive distribution.

28 PDF pp.59,63,70-82; supplementary centered-rank conditions.

29 PDF pp.59,63-64,70-82; supplementary numerical sample.

30 PDF pp.38-44,47-53,88-90; supplementary numerical model comparison.